

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

03

10 answers · Rationale-rich · Teach from doc

Q1**-9**

The correct answer is -9. Since w is in the denominator of a fraction in the given equation, w can't be equal to 0. Since w isn't equal to 0, multiplying both sides of the given equation by w yields an equivalent equation, $-54 = 6w$. Dividing both sides of this equation by 6 yields $-9 = w$. Therefore, -9 is the solution to the given equation.

Q2**C****C. 16**

Choice C is correct. Since the left-hand side of the given equation has a factor of $x + 9$ in both the numerator and the denominator, the solution to the given equation can be found by solving the equation $x - 9 = 7$. Adding 9 to both sides of this equation yields $x = 16$. Substituting 16 for x in the given equation yields $\frac{16 + 916 - 9}{16 + 9} = 7$, or $7 = 7$. Therefore, the solution to the given equation is 16.

Choice A is incorrect. Substituting 7 for x in the given equation yields $\frac{7 + 97 - 9}{7 + 9} = 7$, or $-2 = 7$, which is false.

Choice B is incorrect. Substituting 9 for x in the given equation yields $\frac{9 + 99 - 9}{9 + 9} = 7$, or $0 = 7$, which is false.

Choice D is incorrect. Substituting 63 for x in the given equation yields $\frac{63 + 963 - 9}{63 + 9} = 7$, or $54 = 7$, which is false.

Q3**D**

D. $T = \frac{v - 331.3}{0.606}$

Choice D is correct. To express T in terms of v, subtract 331.3 from both sides of the equation, which gives $v - 331.3 = 0.606T$. Dividing both sides of the equation by 0.606 gives

$$\frac{v - 331.3}{0.606} = T.$$

Choices A, B, and C are incorrect and are the result of incorrect steps while solving for T.

Q4**A**

A. $p = \frac{2}{3}s$

Choice A is correct. It's given that p represents the period of the ocean wave, so the equation

$$s = \frac{3}{2}p$$

can be solved for p to represent the period of the wave in terms of the speed of the

wave. Multiplying both sides of the equation by the reciprocal of $\frac{3}{2}$ will isolate p. This yields

$$\frac{2}{3}s = \frac{2}{3}\left(\frac{3}{2}p\right),$$

which simplifies to $\frac{2}{3}s = p$. Therefore, $p = \frac{2}{3}s$.

Choices B, C, and D are incorrect and may result from errors made when rearranging the equation to solve for p.

Q5**D**

D. $x = by - 72y$

Choice D is correct. Multiplying both sides of the given equation by y yields $y(b - 72) = x$. Distributing on the left-hand side of this equation yields $by - 72y = x$, or $x = by - 72y$. Therefore, the equation $x = by - 72y$ correctly expresses x in terms of b and y .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6**D**

D. Zero

Choice D is correct. Since the square of a real number is never negative, the given equation isn't true for any real value of x . Therefore, the given equation has zero distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

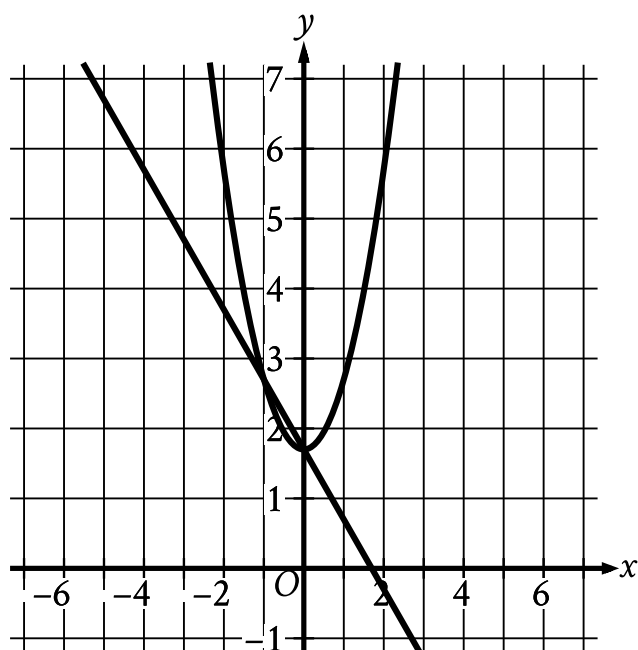
Q7**1**

The correct answer is 1. The point of intersection of the graphs of the given equations is the solution to the system of the two equations. Since $y = (x - 1)^2$ and $y = 2x - 3$, it follows that $(x - 1)^2 = 2x - 3$, or $(x - 1)(x - 1) = 2x - 3$. Applying the distributive property to the left-hand side of this equation yields $x^2 - 2x + 1 = 2x - 3$. Subtracting $2x$ from and adding 3 to both sides of this equation yields $x^2 - 4x + 4 = 0$. Factoring the left-hand side of this equation yields $(x - 2)(x - 2) = 0$. By the zero product property, if $(x - 2)(x - 2) = 0$, it follows that $x - 2 = 0$. Adding 2 to both sides of $x - 2 = 0$ yields $x = 2$. Substituting 2 for x in either of the given equations yields $y = 1$. For example, substituting 2 for x in the second given equation yields $y = 2(2) - 3$, or $y = 1$. Therefore, the point of intersection of the graphs of the given equations is $(2, 1)$. The y -coordinate of this point is 1.

Q8**2**

Accept also: -12

The correct answer is either 2 or -12. The left-hand side of the given equation can be rewritten by factoring. The two values that multiply to -24 and add to 10 are 12 and -2. It follows that the given equation can be rewritten as $z^2 + 12z - 2 = 0$. Setting each factor equal to 0 yields two equations: $z + 12 = 0$ and $z - 2 = 0$. Subtracting 12 from both sides of the equation $z + 12 = 0$ results in $z = -12$. Adding 2 to both sides of the equation $z - 2 = 0$ results in $z = 2$. Note that 2 and -12 are examples of ways to enter a correct answer.

Q9**A****A.**

- For the line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (negative 1 comma 2.7)
 - (0 comma 1.7)
- For the parabola in the system:
 - The parabola opens upward.
 - The vertex is at point (0 comma 1.7).
 - The parabola passes through the following points:
 - (negative 1 comma 2.7)
 - (0 comma 1.7)
 - (1 comma 2.7)

Choice A is correct. The graph of a quadratic equation in the form $y = x^2 + c$ has its vertex at $(0, c)$. The first equation in the given system of equations is $y = x^2 + 1.7$, so the graph of this quadratic equation has its vertex at $(0, 1.7)$. The graph of a linear equation of the form $y = b - x$ has a slope of -1 and a y-intercept at $(0, b)$. The second equation in the given system of equations is $y = 1.7 - x$, so the graph of this linear equation has a slope of -1 and a y-intercept at

$(0, 1.7)$. Of the choices, only choice A has the graph of a quadratic equation with its vertex at $(0, 1.7)$ and the graph of a linear equation with a slope of -1 and a y-intercept at $(0, 1.7)$.

Choice B is incorrect. This graph represents a system in which the second equation is $y = 1.7 + x$, not $y = 1.7 - x$.

Choice C is incorrect. This graph represents a system in which the first equation is $y = (x + 1.7)^2$, not $y = x^2 + 1.7$.

Choice D is incorrect. This graph represents a system in which the first equation is $y = (x + 1.7)^2$, not $y = x^2 + 1.7$, and the second equation is $y = 1.7 + x$, not $y = 1.7 - x$.

Q10

C

C. $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$

Choice C is correct. Using the quadratic formula to solve the given expression yields

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - (4)(1)(-1)}}{(2)(1)} = \frac{1 \pm \sqrt{5}}{2}. \text{ Therefore, } x = \frac{1+\sqrt{5}}{2} \text{ and } x = \frac{1-\sqrt{5}}{2}$$

satisfy the given equation.

Choices A and B are incorrect and may result from incorrectly factoring or incorrectly applying the quadratic formula. Choice D is incorrect and may result from a sign error.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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